

Overview

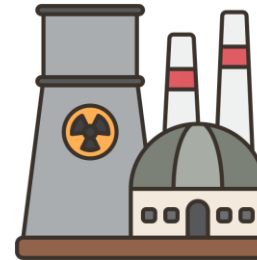
Fundamentals of System Security
(COMSM0122)

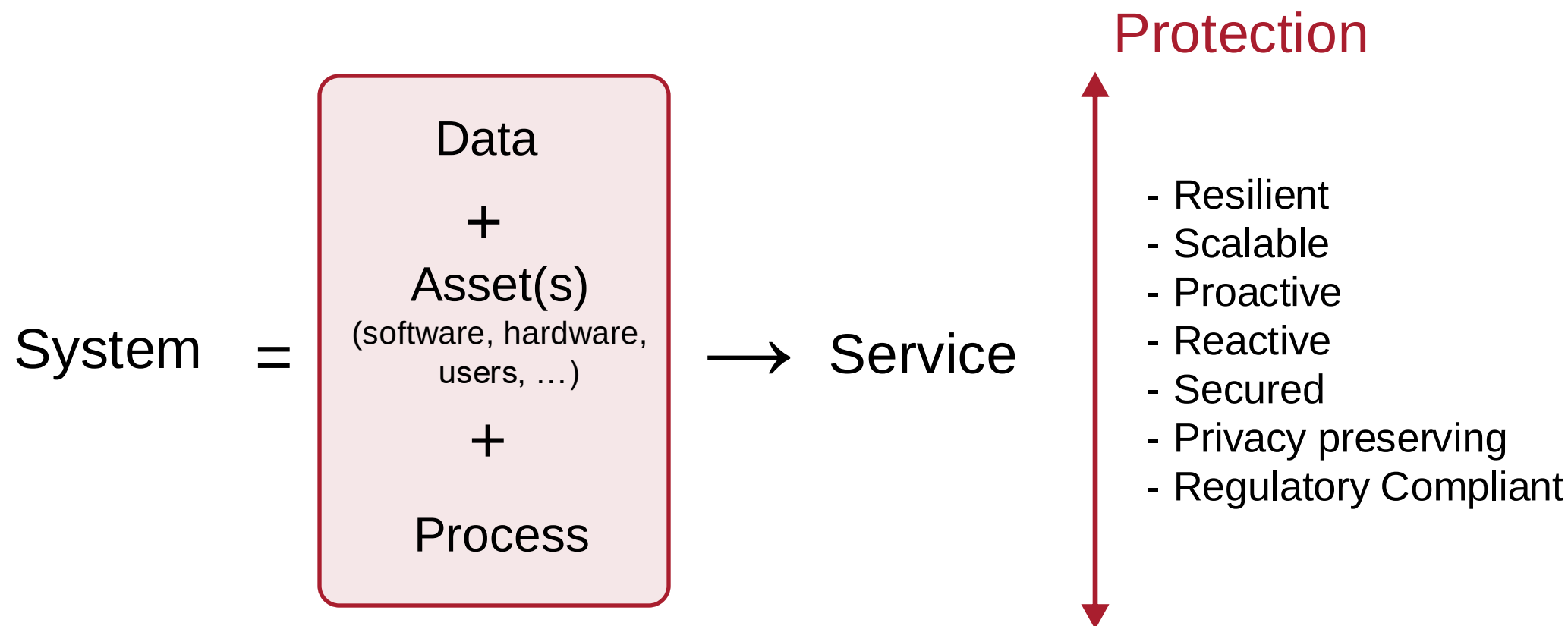
Your learning on this unit

1. Have knowledge of a typical computing system in terms of fundamental/basic components and interactions among them to understand the practical low-level details.
2. Have knowledge about security aspects of software and hardware, including classical and contemporary security vulnerabilities and their impact (exploitation).
3. Be able to critically analyse software and systems design from a security perspective.
4. Be able to demonstrate the understanding of how systems and software can be engineered to protect against offensive techniques.
5. Hands-on knowledge and experience of working on system security in real-world contexts.
6. Learn to communicate about complex technical topics.

Aims

- The aim of this course is to empower you with the techniques and skills needed to ensure the **protection of a computer based systems**.





The interplay between data, asset and process

Systems Security is the study of tools, methods and techniques for building and maintaining services (inc. its infrastructure) to ensure that system's data, assets and processes are protected.

Nature of the Problem

- Modern systems engineering practices are characterized by decentralized and bottom-up decision making
 - Agile
 - Microservices
 - Tech Stack
 - Repositories
 - DevOps
 - ...
- This engineering approach engenders innovation, but makes it difficult to demonstrate assurance of protection at different layers

Nature of the Problem

Some Systems Engineering challenges:

- An employee or equivalent insider accesses sensitive business or customer data when they should not have.
- A business partner obtains business or customer data at a time or in a volume that compromises the customers or the competitive advantage of the business.
- Data that was collected for a benign, defensible purpose gets used for something more than it was intended for.
- A system depends on external APIs without understanding of associated liabilities.
- System is unable to meet its Service Level Agreements - e.g. on down time, scalability, etc.
- Systems engineers are unable to determine the level of protection that the system provides to its users
- ...

Topics

- *Data Protection* (ILO3, 4, 5, 6):
Data Governance including Data Classification, Data Mapping, and Data Protection Principles
- *Process Protection* (ILO1, 4, 6):
System Design, Authentication, Authorisation and Access Control
- *Asset Protection* (ILO1, 2, 3, 6):
Software - Static analysis of software artefacts, including program code, repositories, etc, Malware and bug detection systems.
Infrastructure (Hardware) – Memory Analysis, Cloud Architectures, Virtualisation, High Availability, Scalability, Self-Healing